



Neuralink competitors

Neuralink is actually pretty late to the BCI party, here are the competing companies that you want to track. <https://digit.in/aug19-87>



The market prospects

The global market for brain computer interfaces is projected to reach 2.67 Billion by 2026. <https://digit.in/aug19-88>

whether or not to rotate the block to complete the line.

Five groups of participants played 16 rounds of the game each. During this process, all three participants of a group were in different rooms altogether and couldn't hear, see or speak to one another. Senders see the line and the block along with a prompt asking them if the block should be moved. One side of the screen says 'YES' while the other says 'NO'. Beneath the 'YES' option, an LED flashes 17 times per second and beneath the 'NO' option, the LED flashes only 15 times a second. When the sender makes a decision, they are required to concentrate on the corresponding light. The EEG caps pick up the senders electrical activity, which is generated by the LED lights' different flashing patterns, that trigger unique activity in the brain. The computer provides real-time feedback by displaying a cursor on the screen that moves towards each sender's desired choice. These selections are then translated into a YES or a NO that is received by the 'Receiver'. The team used a cable that ends with a wand and looks like a tiny racket to deliver the message to the receiver. This 'coil' is placed behind the receiver's head and it stimulates the occipital cortex,

as mentioned above, which is the part that translates signals from the eyes.

Stocco said, "We essentially 'trick' the neurons in the back of the brain to spread around the message that they have received from the eyes. Then participants have the sensation that bright arcs or objects suddenly appear in front of their eyes". Simply put, this means, if the answer is 'Yes, rotate the block', the receiver will see a bright flash. If the answer is 'No' then they see nothing. The Receiver also wears an EEG cap which allows him to tell the game his decision to rotate the block or not in the same way. The senders also got a chance to review the decision taken by the receiver and send corrections if needed. Once the receiver inputs the second and final decision, everyone in the group finds out if they managed to clear the line. After rigorously testing five groups of threes, the scientists found that this brain-to-brain interface could achieve an 81 per cent accuracy in the game, which means the teams cleared the game 13 out of 16 times on average.

'GOOD' AND 'BAD' TEAMMATES

The researchers also wanted to delve deeper into whether the Receiver would learn to trust one Sender over the other over time, based on their

reliability. Therefore, the team purposely picked one Sender to be the 'bad Sender'. No, they didn't make the senders answer incorrectly but they just flipped the bad sender's responses in 10 out of 16 trials. The study indicated that the receiver went from being relatively neutral about both senders to strongly preferring information received from one of them, the 'good sender'. This 'noise' essentially allowed the team to observe

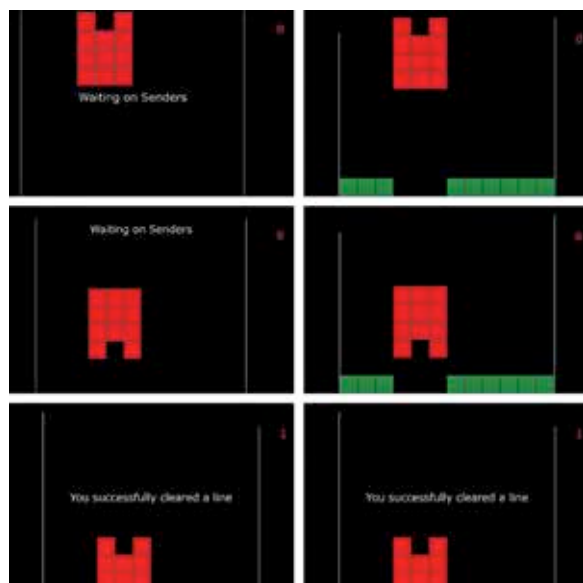
how the Receiver processes the noisy signals in order to make the right decision. So, just like in conventional social networks, the receiver (like a consumer) would learn to trust the sender that provided more reliable information, essentially teasing out fake news in this brain-to-brain social network. Researchers hope that they can use these findings to improve cooperative problem-solving by using a sort of 'social network' of connected brains.



Linxing Preston, a part of the BrainNet team, sets up a 'Sender' for the experiment

THE FUTURE OF COMMUNICATION?

Despite this brain-to-brain system only being able to communicate binary information (YES or NO), efforts are being undertaken to broaden the bandwidth by exploring the usage of functional MRI (fMRI) to read changes in brain activity better. As non-invasive neurotechnologies get better with time, BrainNet will get more sophisticated as well. However, the researchers in the BrainNet team believe that this is the right time to begin conversing about the ethics of this kind of brain augmentation research and start developing protocols to ensure privacy is respected as the technology continues to evolve and get better. The group has begun working with the Neuroethics team at the Centre for Neurotechnology. The lead neuro-engineer, Rao, believes that BrainNet is just the beginning and they have a long way to go since their equipment is still expensive and bulky, and the task is actually a game. The pursuit of these kinds of brain-to-brain interfaces has the potential to open up new frontiers in communication between humans. 



Senders can see the block and the line, but receivers can only see the block